

Mathematical Research Log

Focus: Elliptic Curves & Galois Theory

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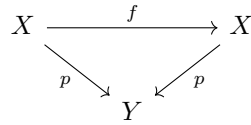
Daily Insight / Philosophy

As this truth is already woven into the fabric of the universe, it belongs to the commons. To have heard it is to possess the right to discover it for myself. I am not stealing; I am re-envisioning it through my own lens. If you only commit a proof to memory, it is theft; but if, through study, you reconstruct its logic until it is as effortless as breath, it becomes part of your very being.

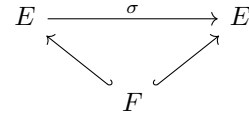
1. Some ideas on the Galois theory of covering spaces

In the Galois theory, for a Galois extension(at least finite) of fields, E/F , a Galois group $\text{Gal}(E/F)$ can be assigned to this extension.

For the covering space theory, a covering X to Y between two topological space is what a Galois extension E to F . Similarly, there should be a group 'corresponds' to the covering $p : X \rightarrow Y$, that is $\text{Aut}(X/Y)$. This group fixes the map working on Y and 'permutes' the cover X . It consists of all Deck-transformation f which makes the following diagram commutes:



(a) Deck transformation



(b) Galois group transformation

Figure 1: Deck transformation and Galois groups

Fix a point $x \in X$. For the covering space, the covering map $p : X \rightarrow Y$ induces a group homomorphism

$$p^* : \pi_n(X, x) \rightarrow \pi_n(Y, p(x)) \quad [\gamma] \mapsto [p \circ \gamma]$$

Similarly, I am curious about the counterpart for $\pi_n(X, x)$ or $\pi_n(Y, y)$ for some $x \in X$ or $y \in Y$ in the Galois theory. Is there a concept called 'fundamental group' in Galois theory?

Let's try to get one for $\pi_n(X, x)$. Since X is the counterpart for E , the group in the Galois theory should be of the form $\pi_n(E, e)$ for some $e \in E$, even though this is not well-defined. To define it, it is inevitable to touch 'a curve' $\gamma : I \rightarrow E$. But, it is impossible to consider the continuity because a general field does not equip a topology on it. To make the concept alive in the Galois theory, another perspective is necessary.

The point of relative enlightens us: When one path leads to a cul-de-sac. Do not always stare an object itself. Just look at how it interacts with any other objects.

Based on this point of view, what objects act on/interact with $\pi_n(X, x)$. Actually, as we just mentioned, $\text{Aut}(X/Y)$ is a candidate. What is their exact relation?

In short,

$$\text{Aut}(X/Y) \cong \pi_n(Y, p(x)) / p^* \pi_n(X, x)$$

If one can choose X to be some X' with $p' : X' \rightarrow Y$ such that $\ker \pi'^* = \pi_n(X', x)$, then for this X' and p' ,

$$\text{Aut}(X'/Y) \cong \pi_n(Y, p(x))$$

Remark A higher perspective explanation of what I stated is: *Instead of defining loops via paths, we define loops via the automorphisms of the objects they classify.*

The objects means those coverings $p : X \rightarrow Y$. The fundamental groups $\pi_1(X, x)$ classify those coverings from the following equivalence of category:

$$\{\text{subgroups } H \leq \pi_1(Y, y)\} \longleftrightarrow \{\text{The coverings } p : (X, x) \rightarrow (Y, y)\}$$

where x satisfies $p(x) = y$.

Once we know the subgroup, it is equivalent to know the corresponding covering space. The maximal subgroup $\pi_1(Y, y)$ corresponds to the universal covering space $\tilde{Y}$ of Y .

The universal cover have some sense of maximality. Translating this maximality to the field extensions, it should

corresponds to some 'closure', like algebraic closure, normal closure or separable closure. Which one should we choose among all these types?

To answer this question, let's look at what property the covering spaces are in possession of. The universal covering $\tilde{X}$ is a simply-connected space, meaning all loops can contract to one single point.

2. Technical Lemma: Local Anatomy of Frey Curves

Recall the definition of the Frey curve $E_{a,b,c} : y^2 = x(x-a)(x+b)$. The discriminant is:

- $\Delta = 16(abc)^2$
- j -invariant calculation...

3. Computation Notes (Python/Mathematica)

```
1 # Finding torsion points on E
2 def check_torsion(a, b):
3     # Mathematica-style logic in Python
4     pass
```

To-Do List

- ☐ Finish the report on bilinear pairings (Tate/Weil).
- ☐ Review the email from Prof. Kamnitzer regarding the semester project.
- ☐ Update GitHub Pages with today's notes.